

unl-statistics / business-report-TylerBrown123

🔍

+

<> Code

⌚ Issues

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▶ Actions

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📈 Insights

Commit 871e2d1

TylerBrown123 committed 5 minutes ago

some changes and presentation slides almost done

[main](#)

1 parent [11ece3a](#) commit 871e2d1

12 files changed +1,824 -377

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- .RData
- .Rhistory
- car_image.jpg
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 - intro.html
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 - intro.qmd
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 - presentation_slides.html
 - presentation_slides.qmd
 - results.rmarkdown

▼ .RData 775 KB

Binary file not shown.

▼ .Rhistory

...

```

...   @@ -1,90 +1,4 @@
1     - "Tornado Path Width (Yards)"
2     - ),
3     - Type = c("Integer", "Date", "Categorical", "Integer", "Numeric", "Numeric", "Numeric",
4         "Numeric", "Numeric", "Numeric"),
5     - Possible_Values = c("1950-2024", "01-01-1950 to 12/31/2024", "Any US State", "0-5 or -9
6         Meaning Unknown", "US Latitudes ~ 29 to 50", "US Longitudes ~ -125 to -70", "US Latitudes ~
7         29 to 50", "US Longitudes ~ -125 to -70", "0-250 Miles", "0-3500 Yards")
8     - )
9     - tornado_vars %>%
10    ☐ gt() %>%
11    - tab_header(
12    - title = "Tornado Data Variable Descriptions"
13    - ) %>%
14    - cols_label(
15    - Variable = "Variable Name",
16    - Description = "Description",
17    - Type = "Type",
18    - Possible_Values = "Possible Values"
19    - ) %>%
20    - tab_style(
21    - style = cell_text(weight = "bold"),
22    - locations = cells_column_labels(everything())
23    - ) %>%
24    - tab_style(
25    - style = cell_borders(
26    - sides = "all",
27    - color = "black",
28    - weight = px(1)
29    - ),
30    - locations = cells_body(everything())
31    - ) %>%
32    - tab_style(
33    - style = cell_borders(
34    - sides = "all",
35    - color = "black",
36    - weight = px(1)
37    - ),
38    - locations = cells_column_labels(everything())
39    - )
40    - library(gt)
41    - tornado_vars <- data.frame(
42    - Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),

```

```

40     - Description = c(
41     - "Year Tornado Occurred",
42     - "Exact Date Tornado Occurred",
43     - "State Abbreviation Where Tornado First Touched Down",
44     - "EF Scale Magnitude of the Tornado",
45     - "Latitude Where the Tornado First Touched Down",
46     - "Longitude Where the Tornado First Touched Down",
47     - "Latitude Where the Tornado Ended",
48     - "Longitude Where the Tornado Ended",
49     - "Tornado Path Length (Miles)",
50     - "Tornado Path Width (Yards)"
51     - ),
52     - Type = c("Integer", "Date", "Categorical", "Integer", "Numeric", "Numeric", "Numeric",
53               "Numeric", "Numeric", "Numeric"),
54     - Possible_Values = c("1950-2024", "01-01-1950 to 12/31/2024", "Any US State", "0-5 or -9
55                           Meaning Unknown", "US Latitudes ~ 29 to 50", "US Longitudes ~ -125 to -70", "US Latitudes ~
56                           29 to 50", "US Longitudes ~ -125 to -70", "0-250 Miles", "0-3500 Yards")
57     - )
58     - tornado_vars %>%
59     - gt() %>%
60     - tab_header(
61     - title = "Tornado Data Variable Descriptions"
62     - ) %>%
63     - cols_label(
64     - Variable = "Variable Name",
65     - Description = "Description",
66     - Type = "Type",
67     - Possible_Values = "Possible Values"
68     - ) %>%
69     - tab_style(
70     - style = cell_text(weight = "bold"),
71     - locations = cells_column_labels(everything())
72     - ) %>%
73     - tab_style(
74     - style = cell_borders(
75     - sides = "all",
76     - color = "black",
77     - weight = px(1)
78     - ),
79     - locations = cells_body(everything())
80     - ) %>%
81     - tab_style(
82     - style = cell_borders(
83     - sides = "all",
84     - color = "black",
85     - weight = px(1)

```

```

83     - ),
84     - locations = cells_column_labels(everything())
85     - )
86     - # Data Cleaning (1 row per tornado, only important columns, dealing with 0's in elat and
      elon)
87     - library(tidyverse)
      1 + library(maps)
88     2   tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
89     3   filter(!st %in% c("AK", "HI", "PR", "VI"))
90     4   tornado_paths <- tornado_full_data |>
      ⚡ @@ -93,122 +7,36 @@ tornado_paths <- tornado_paths |>
93     7   select(slat, slon, elat, elon, mag)
94     8   tornado_paths <- tornado_paths |>
95     9   mutate(end_missing = (elat == 0 | elon == 0))
96     - ggplot(tornado_paths, aes(x = slon, y = slat)) +
97     - geom_point(color = "red", alpha = 0.5) +
      10 + us_states <- map_data("state")
      11 + ggplot() +
      12 + # State borders
      13 + geom_polygon(data = us_states,
      14 + aes(x = long, y = lat, group = group),
      15 + fill = NA, color = "black", linewidth = 0.3) +
      16 + # Tornado points
      17 + geom_point(data = tornado_paths,
      18 + aes(x = slon, y = slat),
      19 + color = "red", alpha = 0.5, size = 0.05) +
98     20   labs(title = "Starting Location For All Tornadoes",
99     - x = "Starting Longitude", y = "Starting Latitude") +
100     - theme_minimal()
101     - ggplot(tornado_paths |> filter(elat == 0 | elon == 0),
102     - aes(x = slon, y = slat)) +
103     - geom_point(color = "red", alpha = 0.5) +
104     - labs(title = "Starting Location For Tornadoes With Missing End Coordinates",
105     - x = "Starting Longitude", y = "Starting Latitude") +
      21 + x = "Longitude", y = "Latitude") +
106     22   theme_minimal()
107     23   library(ggtext)
108     - ggplot(
109     - tornado_paths |>
110     - filter(elat == 0 | elon == 0),
111     - aes(x = slon, y = slat)
112     - ) +
113     - geom_point(color = "red", alpha = 0.5) +
114     - facet_wrap(~ mag) +
      24 + ggplot() +
      25 + geom_polygon(data = us_states,

```

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26 + aes(x = long, y = lat, group = group),
27 + fill = "gray95", color = "black", linewidth = 0.3) +
28 + geom_point(data = tornado_paths |>
29 + filter(mag != -9),
30 + aes(x = slon, y = slat),
31 + color = "red",
32 + alpha = 0.5,
33 + size = 0.05) +
34 + facet_wrap(~ mag, nrow = 2, ncol = 3) +
115 35   labs(
116   - title = "Starting Location For Tornadoes With Missing End Coordinates (by Magnitude)",
117   - x = "Starting Longitude",
118   - y = "Starting Latitude"
119   - ) +
120   - theme_minimal() +
121   - theme(
122   - plot.title = element_markdown()
123   - )
124   - # Transforming latitude and longitude into math ready quantities
125   - library(rnaturalearth)
126   - library(sf)
127   - tornado_paths <- tornado_paths |>
128   - filter(end_missing == FALSE)
129   - tornado_paths$start_geom_degrees <- st_as_sf(
130   - tornado_paths,
131   - coords = c("slon", "slat"),
132   - crs = 4269
133   - )$geometry
134   - tornado_paths$end_geom_degrees <- st_as_sf(
135   - tornado_paths,
136   - coords = c("elon", "elat"),
137   - crs = 4269
138   - )$geometry
139   - tornado_paths$start_geom_meters <- st_transform(
140   - tornado_paths$start_geom_degrees, 5070
141   - )
142   - tornado_paths$end_geom_meters <- st_transform(
143   - tornado_paths$end_geom_degrees, 5070
144   - )
145   - coast_m <- ne_coastline(returnclass = "sf") |>
146   - st_transform(5070)
147   - # Find distances to coast and classify each tornado as coastal_to_inland , inland_to_coastal
    , or other
148   - tornado_paths <- tornado_paths |>
149   - mutate(
150   - dist_start_to_coast = st_distance(start_geom_meters, coast_m) |> apply(1, min),

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151   - dist_end_to_coast = st_distance(end_geom_meters, coast_m) |> apply(1, min)
152   - )
153   - tornado_paths <- tornado_paths |>
154   - mutate(
155   -   movement = case_when(
156   -     dist_start_to_coast <= 10000 & dist_end_to_coast > dist_start_to_coast ~ "Coastal_to_Inland",
157   -     dist_start_to_coast > 10000 & dist_end_to_coast < dist_start_to_coast ~ "Inland_to_Coastal",
158   -     TRUE ~ "Other"
159   -   )
160   - )
161   - library(gt)
162   - tornado_vars <- data.frame(
163   -   Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),
164   -   Description = c(
165   -     "Year Tornado Occurred",
166   -     "Exact Date Tornado Occurred",
167   -     "State Abbreviation Where Tornado First Touched Down",
168   -     "EF Scale Magnitude of the Tornado",
169   -     "Latitude Where the Tornado First Touched Down",
170   -     "Longitude Where the Tornado First Touched Down",
171   -     "Latitude Where the Tornado Ended",
172   -     "Longitude Where the Tornado Ended",
173   -     "Tornado Path Length (Miles)",
174   -     "Tornado Path Width (Yards)"
175   -   ),
176   -   Type = c("Integer", "Date", "Categorical", "Integer", "Numeric", "Numeric", "Numeric",
177   -     "Numeric", "Numeric", "Numeric"),
178   -   Possible_Values = c("1950-2024", "01-01-1950 to 12/31/2024", "Any US State", "0-5 or -9
179   -     Meaning Unknown", "US Latitudes ~ 29 to 50", "US Longitudes ~ -125 to -70", "US Latitudes ~
180   -     29 to 50", "US Longitudes ~ -125 to -70", "0-250 Miles", "0-3500 Yards")
181   -   + title = "Starting Location All Tornadoes Grouped by Magnitude",
182   -   + x = "Longitude",
183   -   + y = "Latitude"
184   -   )
185   - tornado_vars %>%
186   -   gt() %>%
187   -   tab_header(
188   -     title = "Tornado Data Variable Descriptions"
189   -   ) %>%
190   -   cols_label(
191   -     Variable = "Variable Name",
192   -     Description = "Description",
193   -     Type = "Type",
194   -     Possible_Values = "Possible Values"
195   -   ) %>%
196   -   tab_style(

```

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191     - style = cell_text(weight = "bold"),
192     - locations = cells_column_labels(everything())
193     - ) %>%
194     - tab_style(
195     - style = cell_borders(
196     - sides = "all",
197     - color = "black",
198     - weight = px(1)
199     - ),
200     - locations = cells_body(everything())
201     - ) %>%
202     - tab_style(
203     - style = cell_borders(
204     - sides = "all",
205     - color = "black",
206     - weight = px(1)
207     - ),
208     - locations = cells_column_labels(everything())
209     - )
210     - # Data Cleaning (1 row per tornado, only important columns, dealing with 0's in elat and
      elon)
211     - library(tidyverse)
212 40     tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
213 41     filter(!st %in% c("AK", "HI", "PR", "VI"))
214 42     tornado_paths <- tornado_full_data |>
      ⚡@@ -217,12 +45,37 @@ tornado_paths <- tornado_paths |>
217 45     select(slat, slon, elat, elon, mag)
218 46     tornado_paths <- tornado_paths |>
219 47     mutate(end_missing = (elat == 0 | elon == 0))
220     - ggplot(tornado_paths, aes(x = slon, y = slat)) +
221     - geom_point(color = "red", alpha = 0.5) +
      48 + us_states <- map_data("state")
      49 + ggplot() +
      50 + geom_polygon(data = us_states,
      51 + aes(x = long, y = lat, group = group),
      52 + fill = NA, color = "black", linewidth = 0.3) +
      53 + geom_point(data = tornado_paths,
      54 + aes(x = slon, y = slat),
      55 + color = "red", alpha = 0.5, size = 0.05) +
222 56     labs(title = "Starting Location For All Tornadoes",
223     - x = "Starting Longitude", y = "Starting Latitude") +
      57 + x = "Longitude", y = "Latitude") +
224 58     theme_minimal()
225     - {r, echo = FALSE}
      59 + tornado_full_data
      60 + tornado_data |>

```

```

61 + filter(mag = 4)
62 + tornado_data |>
63 + filter(mag == 4)
64 + tornado_paths |>
65 + filter(mag == 4)
66 + tornado_paths |>
67 + filter(mag == 4 & movement = coastal_to_inland)
68 + tornado_paths |>
69 + filter(mag == 4 & movement == coastal_to_inland)
70 + tornado_paths
71 + # Find distances to coast and classify each tornado as coastal_to_inland , inland_to_coastal
    , or other
72 + tornado_paths <- tornado_paths |>
73 + mutate(
74 + dist_start_to_coast = st_distance(start_geom_meters, coast_m) |> apply(1, min),
75 + dist_end_to_coast = st_distance(end_geom_meters, coast_m) |> apply(1, min)
76 + )
77 + library(tidyverse)
78 + library(maps)
226 79   library(gt)
227 80   tornado_vars <- data.frame(
228 81     Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),
    ↓
    ↑
272 125   ),
273 126   locations = cells_column_labels(everything())
274 127   )
    128 + #| label: fig-tornado-all
    129 + #| fig-cap: "Starting locations of all tornadoes in the continental United States."
275 130   # Data Cleaning (1 row per tornado, only important columns, dealing with 0's in elat and
    elon)
276   - library(tidyverse)
277 131   tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
278 132   filter(!st %in% c("AK", "HI", "PR", "VI"))
279 133   tornado_paths <- tornado_full_data |>
    ↕
282 136   select(slat, slon, elat, elon, mag)
283 137   tornado_paths <- tornado_paths |>
284 138   mutate(end_missing = (elat == 0 | elon == 0))
285   - ggplot(tornado_paths, aes(x = slon, y = slat)) +
286   - geom_point(color = "red", alpha = 0.5) +
    139 + us_states <- map_data("state")
    140 + ggplot() +
    141 + geom_polygon(data = us_states,
    142 + aes(x = long, y = lat, group = group),
    143 + fill = NA, color = "black", linewidth = 0.3) +

```



```

144 + geom_point(data = tornado_paths,
145 + aes(x = slon, y = slat),
146 + color = "red", alpha = 0.5, size = 0.05) +
287 147   labs(title = "Starting Location For All Tornadoes",
288       - x = "Starting Longitude", y = "Starting Latitude") +
148 + x = "Longitude", y = "Latitude") +
289 149   theme_minimal()
290       - ggplot(tornado_paths |> filter(elat == 0 | elon == 0),
291       - aes(x = slon, y = slat)) +
292       - geom_point(color = "red", alpha = 0.5) +
293       - labs(title = "Starting Location For Tornadoes With Missing End Coordinates",
294       - x = "Starting Longitude", y = "Starting Latitude") +
150 + #| label: fig-tornado-missingend
151 + #| fig-cap: "Starting locations of tornadoes with missing end coordinates."
152 + ggplot() +
153 + geom_polygon(data = us_states,
154 + aes(x = long, y = lat, group = group),
155 + fill = NA, color = "black", linewidth = 0.3) +
156 + geom_point(data = tornado_paths |>
157 + filter(elat == 0 | elon == 0),
158 + aes(x = slon, y = slat),
159 + color = "red",
160 + alpha = 0.5,
161 + size = 0.05) +
162 + labs(title = "Starting Locations of Tornadoes with Missing End Coordinates",
163 + x = "Longitude", y = "Latitude") +
295 164   theme_minimal()
165 + #| label: fig-tornado-allmagnitude
166 + #| fig-cap: "Starting locations of all tornadoes in the continental United States grouped by
      magnitude on EF scale"
296 167   library(ggtext)
297       - ggplot(
298       - tornado_paths |>
168 + ggplot() +
169 + geom_polygon(data = us_states,
170 + aes(x = long, y = lat, group = group),
171 + fill = "gray95", color = "black", linewidth = 0.3) +
172 + geom_point(data = tornado_paths |>
173 + filter(mag != -9),
174 + aes(x = slon, y = slat),
175 + color = "red",
176 + alpha = 0.5,
177 + size = 0.05) +
178 + facet_wrap(~ mag) +
179 + labs(
180 + title = "Starting Location All Tornadoes Grouped by Magnitude",

```

```

181 + x = "Longitude",
182 + y = "Latitude"
183 + )
184 + #| label: fig-tornado-allmagnitudemissingend
185 + #| fig-cap: "Starting locations of tornadoes with missing end coordinates grouped by
      magnitude on EF scale"
186 + library(ggtext)
187 + ggplot() +
188 + geom_polygon(data = us_states,
189 + aes(x = long, y = lat, group = group),
190 + fill = "gray95", color = "black", linewidth = 0.3) +
191 + geom_point(data = tornado_paths |>
299 192   filter(elat == 0 | elon == 0),
300   - aes(x = slon, y = slat)
301   - ) +
302   - geom_point(color = "red", alpha = 0.5) +
      193 + aes(x = slon, y = slat),
      194 + color = "red",
      195 + alpha = 0.5,
      196 + size = 0.1) +
303 197   facet_wrap(~ mag) +
304 198   labs(
305     - title = "Starting Location For Tornadoes With Missing End Coordinates (by Magnitude)",
306     - x = "Starting Longitude",
307     - y = "Starting Latitude"
308     - ) +
309     - theme_minimal() +
310     - theme(
311     - plot.title = element_markdown()
      199 + title = "Starting Location for Tornadoes with Missing End Coordinates Grouped by Magnitude",
      200 + x = "Longitude",
      201 + y = "Latitude"
312 202   )
313 203   # Transforming latitude and longitude into math ready quantities
314 204   library(rnaturalearth)
      ↓
      @@ -347,6 +237,116 @@ dist_start_to_coast > 10000 & dist_end_to_coast <
      ↑
      dist_start_to_coast ~ "Inland_
347 237   TRUE ~ "Other"
348 238   )
349 239   )
      240 + tornado_paths
      241 + tornado_paths |>
      242 + filter(mag == 4 & movement = Coastal_to_Inland)
      243 + tornado_paths |>
      244 + filter(mag == 4 & movement == Coastal_to_Inland)
      245 + tornado_paths |>

```

```
246 + filter(mag == 4 & movement == "Coastal_to_Inland")
247 + tornado_paths |>
248 + filter(mag == 5 & movement == "Coastal_to_Inland")
249 + tornado_paths |>
250 + filter(mag == 4 & movement == "Coastal_to_Inland")
251 + tornado_data |>
252 + filter(elat == 28.43)
253 + tornado_data |>
254 + filter(elat == 28.43)
255 + tornado_data |>
256 + filter(mag == 4 & elat == 28.43)
257 + tornado_data |>
258 + filter(mag == 4 & elat == 28.43)
259 + tornado_paths |>
260 + filter(mag == 4 & movement == "Coastal_to_Inland")
261 + florida_tornado <- tornado_paths |>
262 + filter(mag == 4 & movement == "Coastal_to_Inland")
263 + florida <- map_data("state") |>
264 + filter(region == "florida")
265 + florida
266 + florida_tornado <- tornado_paths |>
267 + filter(mag == 4 & movement == "Coastal_to_Inland")
268 + florida <- map_data("state") |>
269 + filter(region == "florida")
270 + ggplot() +
271 + geom_polygon(
272 + data = florida,
273 + aes(x = long, y = lat, group = group),
274 + fill = NA,
275 + color = "black",
276 + linewidth = 0.3
277 + ) +
278 + geom_point(
279 + data = fl_tornado,
280 + aes(x = slon, y = slat),
281 + color = "red",
282 + size = 3
283 + ) +
284 + labs(
285 + title = "Only Possible Sharknado Tornado Found",
286 + x = "Longitude",
287 + y = "Latitude"
288 + )
289 + florida_tornado <- tornado_paths |>
290 + filter(mag == 4 & movement == "Coastal_to_Inland")
291 + florida <- map_data("state") |>
```

```
292 + filter(region == "florida")
293 + ggplot() +
294 + geom_polygon(
295 + data = florida,
296 + aes(x = long, y = lat, group = group),
297 + fill = NA,
298 + color = "black",
299 + linewidth = 0.3
300 + ) +
301 + geom_point(
302 + data = florida_tornado,
303 + aes(x = slon, y = slat),
304 + color = "red",
305 + size = 3
306 + ) +
307 + labs(
308 + title = "Only Possible Sharknado Tornado Found",
309 + x = "Longitude",
310 + y = "Latitude"
311 + )
312 + florida_tornado <- tornado_paths |>
313 + filter(mag == 4 & movement == "Coastal_to_Inland")
314 + florida <- map_data("state") |>
315 + filter(region == "florida")
316 + ggplot() +
317 + geom_polygon(
318 + data = florida,
319 + aes(x = long, y = lat, group = group),
320 + fill = NA,
321 + color = "black",
322 + linewidth = 0.3
323 + ) +
324 + geom_point(
325 + data = florida_tornado,
326 + aes(x = slon, y = slat),
327 + color = "red",
328 + size = 3
329 + ) +
330 + geom_point(
331 + data = florida_tornado,
332 + aes(x = elon, y = elat),
333 + color = "blue",
334 + size = 3
335 + ) +
336 + labs(
337 + title = "Only Possible Sharknado Tornado Found",
```

```

338 + x = "Longitude",
339 + y = "Latitude"
340 + )
341 + #Find probabilities of each magnitude of each type of tornado
342 + options(scipen = 999)
343 + probabilities <- tornado_paths |>
344 + group_by(mag, movement) |>
345 + summarise(count = n(), .groups = "drop") |>
346 + group_by(movement) |>
347 + mutate(magnitude_proportion_given_movement_type = count / sum(count)) |>
348 + ungroup() |>
349 + mutate(proportion_of_total_tornadoes = count / total_tornadoes)
350 350   library(gt)
351 351   tornado_vars <- data.frame(
352 352   Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),

```



▼ car_image.jpg

1.8 MB



▼ docs/intro.html



```

↑
@@ -225,7 +225,7 @@ <h1 class="title"><span class="chapter-number">3</span>&nbsp;<span
class="chapt
225 225   </div>
226 226   </div>
227 227   </div>
228     - <p>This project aims to estimate the probability of a sharknado occurring by breaking up the
phenomenon into a sequence of events that need to occur. In the methods section, we outline
the sequence of these probabilities. These include the probability of a tornado forming near

```

228 + <p>This project aims to estimate the probability of a sharknado occurring by breaking up the phenomenon into a sequence of events that need to occur. In the methods section, we outline the sequence of these probabilities. These include the probability of a tornado forming near a coastal region and moving inward, the probability that such a tornado reaches a sufficient intensity to pick up and move a shark, and the assumption that a shark is present at the exact location required. To ensure a conservative upper estimate for this probability, several assumptions are intentionally biased in favor of the event occurring, such as a constant availability of sharks along coastlines and the ability for high magnitude earthquakes to pick them up and transport them inland.</p>

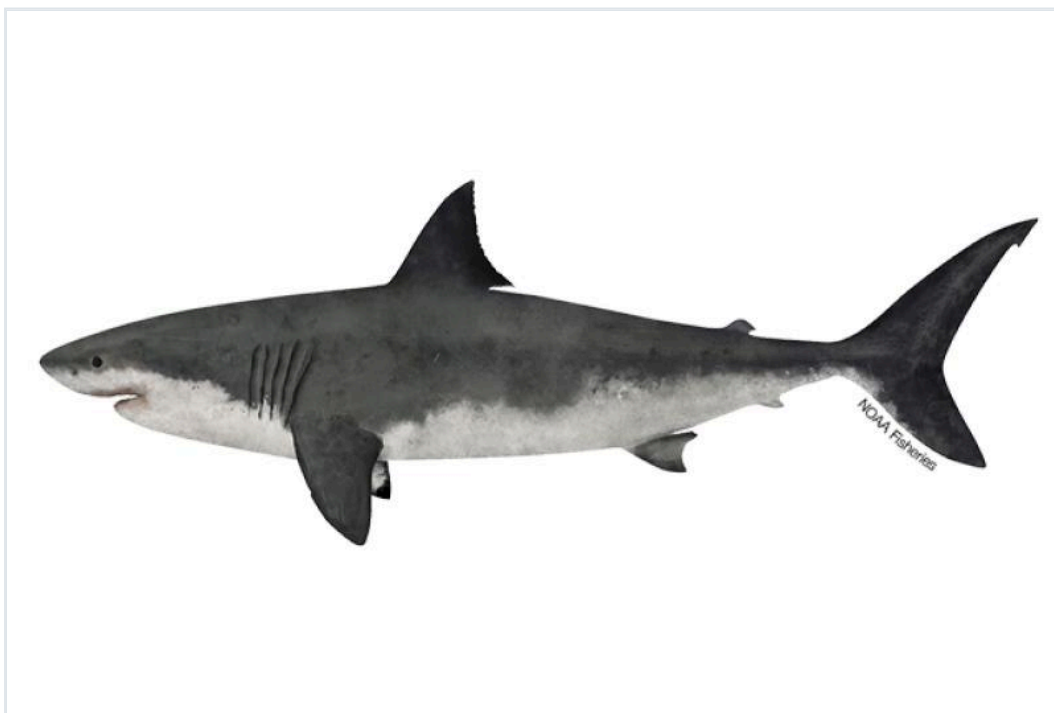
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Large diffs are not rendered by default.

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▼ intro.qmd

...

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↑
... @@ -10,9 +10,6 @@ Maybe make a reference to how it was popularized with the movie? How has
    it piqu
10  10  :::
11  11
12  12  This project aims to estimate the probability of a sharknado occurring by breaking up the
        phenomenon into a sequence of events that need to occur. In the methods section, we outline
        the sequence of these probabilities. These include the probability of a tornado forming near a
        coastal region and moving inward, the probability that such a tornado reaches a sufficient
        intensity to pick up and move a shark, and the assumption that a shark is present at the exact
        location required. To ensure a conservative upper estimate for this probability, several
        assumptions are intentionally biased in favor of the event occurring, such as a constant
        availability of sharks along coastlines and the ability for high magnitude earthquakes to pick
        them up and transport them inland.
13      - ::: callout
14      - fixed spelling on occurring
15      - :::
16  13
17  14  ::: {#fig-sharknadoflowchart}
18  15  ```{mermaid}
↓

```

▼ methods.qmd

...

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↑
... @@ -9,7 +9,9 @@ maybe introduce this section better and the table? Seems outta place,
    like a quo

```

```

9      9
10     10      A table summarizing the tornado data from the NOAA and their possible values is below:
11     11
12     12      - ``{r, echo = FALSE}
13     12      + ``{r, message = FALSE, echo = FALSE}
14     13      + library(tidyverse)
15     14      + library(maps)
16     13     15      library(gt)
17     14     16
18     15     17      tornado_vars <- data.frame(
19     16     17      @@ -65,15 +67,18 @@ tornado_vars %>%
20     17     18      .....
21     18     19      .....
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```

78 + ```{r, echo = FALSE}
79 + #| label: fig-tornado-all
80 + #| fig-cap: "Starting locations of all tornadoes in the continental United States."
75 81   # Data Cleaning (1 row per tornado, only important columns, dealing with 0's in elat and
      elon)
76   - library(tidyverse)
77   82
78   83   tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
79   84   filter(!st %in% c("AK", "HI", "PR", "VI"))
      ⚡@@ -87,56 +92,97 @@ tornado_paths <- tornado_paths |>
87   92   tornado_paths <- tornado_paths |>
88   93   mutate(end_missing = (elat == 0 | elon == 0))
89   94
90   - ggplot(tornado_paths, aes(x = slon, y = slat)) +
91   -   geom_point(color = "red", alpha = 0.5) +
      95 + us_states <- map_data("state")
      96 +
      97 + ggplot() +
      98 +   geom_polygon(data = us_states,
      99 +               aes(x = long, y = lat, group = group),
      100 +               fill = NA, color = "black", linewidth = 0.3) +
      101 +   geom_point(data = tornado_paths,
      102 +               aes(x = slon, y = slat),
      103 +               color = "red", alpha = 0.5, size = 0.05) +
      104 +
      105   labs(title = "Starting Location For All Tornadoes",
82 93     x = "Starting Longitude", y = "Starting Latitude") +
      106 +   x = "Longitude", y = "Latitude") +
83 107   theme_minimal()
84 108
85 109   ```
86 110
87 111
88 112   - ::: callout
89 113   - potentially make the dots smaller and overlay the borders of the US so you can see which are
      on land or not?
90 114   - :::
91 115
92 116   ```{r, echo = FALSE}
93 117   - ggplot(tornado_paths |> filter(elat == 0 | elon == 0),
94 118     aes(x = slon, y = slat)) +
95 119     geom_point(color = "red", alpha = 0.5) +
96 120     labs(title = "Starting Location For Tornadoes With Missing End Coordinates",
97 121       x = "Starting Longitude", y = "Starting Latitude") +
98 122     + #| label: fig-tornado-missingend
99 123     + #| fig-cap: "Starting locations of tornadoes with missing end coordinates."
100 124     + ggplot() +

```

```

116 +   geom_polygon(data = us_states,
117 +               aes(x = long, y = lat, group = group),
118 +               fill = NA, color = "black", linewidth = 0.3) +
119 +   geom_point(data = tornado_paths |>
120 +             filter(elat == 0 | elon == 0),
121 +             aes(x = slon, y = slat),
122 +             color = "red",
123 +             alpha = 0.5,
124 +             size = 0.05) +
125 +
126 +   labs(title = "Starting Locations of Tornadoes with Missing End Coordinates",
127 +        x = "Longitude", y = "Latitude") +
108 128   theme_minimal()
109 129   ```
110 130
111 131   There is no apparent trend in the missing end coordinates. Their geographic distribution is
        similar to that of all tornadoes, suggesting that the first concern is alleviated.
112 132
113      - To address the second concern regarding magnitude, plots can be created for each magnitude
        level to examine whether the incidence of missing end coordinates differs across magnitudes.
133 + To address the second concern regarding magnitude, we can create plots for each magnitude
        level to examine whether the rate of missing end coordinates varies by magnitude, when
        compared to the overall pattern in the full dataset.
114 134
115 135   ```{r, echo = FALSE}
136 + #| label: fig-tornado-allmagnitude
137 + #| fig-cap: "Starting locations of all tornadoes in the continental United States grouped by
        magnitude on EF scale"
116 138   library(ggtext)
117      - ggplot(
118      -   tornado_paths |>
119      -     filter(elat == 0 | elon == 0),
120      -     aes(x = slon, y = slat)
121      - ) +
122      -   geom_point(color = "red", alpha = 0.5) +
139 + ggplot() +
140 +   geom_polygon(data = us_states,
141 +               aes(x = long, y = lat, group = group),
142 +               fill = "gray95", color = "black", linewidth = 0.3) +
143 +   geom_point(data = tornado_paths |>
144 +             filter(mag != -9),
145 +             aes(x = slon, y = slat),
146 +             color = "red",
147 +             alpha = 0.5,
148 +             size = 0.05) +
123 149   facet_wrap(~ mag) +

```

```

150 +
124 151   labs(
125     -   title = "Starting Location For Tornadoes With Missing End Coordinates (by Magnitude)",
126     -   x = "Starting Longitude",
127     -   y = "Starting Latitude"
128     - ) +
129     - theme_minimal() +
130     - theme(
131     -   plot.title = element_markdown()
152 +   title = "Starting Location All Tornadoes Grouped by Magnitude",
153 +   x = "Longitude",
154 +   y = "Latitude"
155 + )
156 + ```
157 +
158 + ```{r, echo = FALSE}
159 + #| label: fig-tornado-allmagnitudemissingend
160 + #| fig-cap: "Starting locations of tornadoes with missing end coordinates grouped by
    magnitude on EF scale"
161 + library(ggtext)
162 + ggplot() +
163 +   geom_polygon(data = us_states,
164 +               aes(x = long, y = lat, group = group),
165 +               fill = "gray95", color = "black", linewidth = 0.3) +
166 +   geom_point(data = tornado_paths |>
167 +             filter(elat == 0 | elon == 0),
168 +             aes(x = slon, y = slat),
169 +             color = "red",
170 +             alpha = 0.5,
171 +             size = 0.1) +
172 +   facet_wrap(~ mag) +
173 +
174 +   labs(
175 +     title = "Starting Location for Tornadoes with Missing End Coordinates Grouped by
    Magnitude",
176 +     x = "Longitude",
177 +     y = "Latitude"
132 178   )
133 179   ```
134 180
135 181   ::: callout-note
136 182   There are no tornadoes with a recorded magnitude of -9 (unknown) that also have missing end
    coordinates.
137 183   :::
138 184

```

```

139      - From these plots, there are no substantial differences in geographic distribution across
      magnitudes. This alleviates the second concern, indicating that removing these tornadoes will
      not meaningfully affect inference on the probability of a sharknado.

185      + From these plots, it appears that higher magnitude tornadoes occur more frequently in the
      Midwest and South and less often on the West Coast. However, this pattern is consistent with
      the plots of all tornadoes separated by magnitude. Therefore, while there is some variation
      by magnitude, excluding tornadoes with missing end coordinates should not heavily bias the
      analysis, as the trend in magnitude are similar between the full dataset and the subset with
      missing end coordinates.

140      186
141      187   ``{r, message = FALSE, echo = FALSE}
142      188   # Transforming latitude and longitude into math ready quantities

      .....
      ↓
      ↑
      .....

181      227   P(\text{sharknado | tornado is coastal to inland}) =
182      228   P(\text{shark is present}) \times
183      229   P(\text{tornado is strong enough to lift a shark | tornado is coastal to inland})
184      - $$ The latter case is likely the better answer to our question, as we are worried about the
      possibility of sharknados occurring when there is a coastal to inland tornado, not when there
      is a tornado anywhere. As such, the following analysis will focus on this conditional
      framework.

      230      + $$

185      231
186      - ::: callout
187      - sharknados? spelling error
188      - :::

      232      + The latter case is likely the better answer to our question, as we are worried about the
      possibility of sharknados occurring when there is a coastal to inland tornado, not when there
      is a tornado anywhere. As such, the following analysis will focus on this conditional
      framework.

189      233
190      234   To obtain a conservative upper bound on this probability, we assume that  $P(\text{shark is
      present}) = 1\$$ . In other words, we assume that whenever a coastal-to-inland tornado occurs, a
      shark is present near the coast and available to be lifted. While this assumption is
      unrealistic, it provides an upper bound on the probability of a sharknado. This strengthens
      the conclusion if the resulting probability remains small.

191      235
      .....
      ↓
      ↑
      .....

222      266   | Other | Could be either | Could be either | Doesn't meet the 2 criteria for either group |
223      267
224      268   ::: {style="text-align:center;"}
      269      +
225      270   
226      - :::
227      271

```

```
228 - <p style="text-align:center;">
272 + <br>
229 273
230 - [Coastal to Inland]{style="color:red;"} \ | [Inland to Coastal]{style="color:blue;"}
274 + <span style="color:red;">Coastal to Inland</span> |
275 + <span style="color:blue;">Inland to Coastal</span>
231 276
232 - </p>
277 + :::
233 278
234 279 ::: callout
235 280 include transition into the next section
236 281
237 282 also, overall would like to see some outside research on methods and why selecting this one
238 283 :::
284 +
285 +
286 +
287 +
288 +
289 +
290 + ```{r}
291 + florida_tornado <- tornado_paths |>
292 +   filter(mag == 4 & movement == "Coastal_to_Inland")
293 +
294 + florida <- map_data("state") |>
295 +   filter(region == "florida")
296 +
297 + ggplot() +
298 +   geom_polygon(
299 +     data = florida,
300 +     aes(x = long, y = lat, group = group),
301 +     fill = NA,
302 +     color = "black",
303 +     linewidth = 0.3
304 +   ) +
305 +   geom_point(
306 +     data = florida_tornado,
307 +     aes(x = slon, y = slat),
308 +     color = "red",
309 +     size = 3
310 +   ) +
311 +   geom_point(
312 +     data = florida_tornado,
313 +     aes(x = elon, y = elat),
314 +     color = "blue",
```

```
315 +   size = 3
316 + ) +
317 + labs(
318 +   title = "Only Possible Sharknado Tornado Found",
319 +   x = "Longitude",
320 +   y = "Latitude"
321 + )
322 + ```
323 +
```

presentation_slides.html

...

Load Diff

Large diffs are not rendered by default.

presentation_slides.qmd

...

```
↑... @@ -4,13 +4,323 @@ author: "Tyler Brown"
4   4   format: revealjs
5   5   ---
6   6
7   7   - ## General Process {.smaller}
8   8   + ## What Data Was Used? {.smaller}
9   9   + - Data from the National Oceanic and Atmospheric Administration (NOAA) on all United States
10  10   +   tornadoes from 1950–2024
11  11   + ```{r, message = FALSE, echo = FALSE}
12  12   + library(gt)
13  13   + library(tidyverse)
14  14   + library(maps)
15  15   +
16  16   + tornado_vars <- data.frame(
17  17   +   Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),
18  18   +   Description = c(
19  19   +     "Year Tornado Occurred",
20  20   +     "Exact Date Tornado Occurred",
21  21   +     "State Abbreviation Where Tornado First Touched Down",
22  22   +     "EF Scale Magnitude of the Tornado",
23  23   +     "Latitude Where the Tornado First Touched Down",
24  24   +     "Longitude Where the Tornado First Touched Down",
25  25   +     "Latitude Where the Tornado Ended",
26  26   +     "Longitude Where the Tornado Ended",
```

```

27 +   "Tornado Path Length (Miles)",
28 +   "Tornado Path Width (Yards)"
29 + ),
30 +   Type = c("Integer", "Date", "Categorical", "Integer", "Numeric", "Numeric", "Numeric",
31 +           "Numeric", "Numeric", "Numeric"),
32 +   Possible_Values = c("1950-2024", "01-01-1950 to 12/31/2024", "Any US State", "0-5 or -9
33 +           Meaning Unknown", "US Latitudes ~ 29 to 50", "US Longitudes ~ -125 to -70", "US Latitudes ~
34 +           29 to 50", "US Longitudes ~ -125 to -70", "0-250 Miles", "0-3500 Yards")
35 + )
36 +
37 + tornado_vars %>%
38 +   gt() %>%
39 +   tab_header(
40 +     title = "Tornado Data Variable Descriptions"
41 +   ) %>%
42 +   cols_label(
43 +     Variable = "Variable Name",
44 +     Description = "Description",
45 +     Type = "Type",
46 +     Possible_Values = "Possible Values"
47 +   ) %>%
48 +   tab_style(
49 +     style = cell_text(weight = "bold"),
50 +     locations = cells_column_labels(everything())
51 +   ) %>%
52 +   tab_style(
53 +     style = cell_borders(
54 +       sides = "all",
55 +       color = "black",
56 +       weight = px(1)
57 +     ),
58 +     locations = cells_body(everything())
59 +   ) %>%
60 +   tab_style(
61 +     style = cell_borders(
62 +       sides = "all",
63 +       color = "black",
64 +       weight = px(1)
65 +     ),
66 +     locations = cells_column_labels(everything())
67 +   )
68 +   ``
69 +
70 + ## Missing Values {.smaller}
71 +

```

```
69 + - NOAA data set has many "0" values for the ending latitudes and longitudes that are really
    NA's
70 +
71 + 
72 +
73 + ## Missing Values cont. {.smaller}
74 +
75 + - Missing values could be a problem if they have a geographical trend
76 +
77 + ::: columns
78 +
79 + ::: column
80 + ```{r, fig.width=7}
81 + tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
82 +   filter(!st %in% c("AK", "HI", "PR", "VI"))
83 +
84 + tornado_paths <- tornado_full_data |>
85 +   distinct(slat, slon, elat, elon, .keep_all = TRUE)
86 +
87 + tornado_paths <- tornado_paths |>
88 +   select(slat, slon, elat, elon, mag)
89 +
90 + tornado_paths <- tornado_paths |>
91 +   mutate(end_missing = (elat == 0 | elon == 0))
92 +
93 + us_states <- map_data("state")
94 +
95 + ggplot() +
96 +   geom_polygon(data = us_states,
97 +               aes(x = long, y = lat, group = group),
98 +               fill = NA, color = "black", linewidth = 0.3) +
99 +   geom_point(data = tornado_paths,
100 +             aes(x = slon, y = slat),
101 +             color = "red", alpha = 0.5, size = 0.05) +
102 +
103 +   labs(title = "Starting Location For All Tornadoes",
104 +        x = "Longitude", y = "Latitude") +
105 +   theme_minimal()
106 +
107 + ```
108 + :::
109 +
110 + ::: column
111 + ```{r fig.width = 7}
112 + ggplot() +
113 +   geom_polygon(data = us_states,
```



```
114 +         aes(x = long, y = lat, group = group),
115 +         fill = NA, color = "black", linewidth = 0.3) +
116 +   geom_point(data = tornado_paths |>
117 +             filter(elat == 0 | elon == 0),
118 +             aes(x = slon, y = slat),
119 +             color = "red",
120 +             alpha = 0.5,
121 +             size = 0.05) +
122 +
123 +   labs(title = "Starting Locations of Tornadoes with Missing End Coordinates",
124 +        x = "Longitude", y = "Latitude") +
125 +   theme_minimal()
126 + ```
127 + :::
128 +
129 + :::
130 +
131 + ## Missing Values cont. {.smaller}
132 +
133 + ::: columns
134 +
135 + ::: column
136 + ```{r, fig.height=12}
137 + library(ggtext)
138 + ggplot() +
139 +   geom_polygon(data = us_states,
140 +               aes(x = long, y = lat, group = group),
141 +               fill = "gray95", color = "black", linewidth = 0.3) +
142 +   geom_point(data = tornado_paths |>
143 +             filter(mag != -9),
144 +             aes(x = slon, y = slat),
145 +             color = "red",
146 +             alpha = 0.5,
147 +             size = 0.4) +
148 +   facet_wrap(~ mag, nrow = 3, ncol = 2) +
149 +
150 +   labs(
151 +     title = "Starting Location All Tornadoes Grouped by Magnitude",
152 +     x = "Longitude",
153 +     y = "Latitude"
154 +   )
155 + ```
156 + :::
157 +
158 + ::: column
159 + ```{r fig.height = 12}
```

```

160 + library(ggtext)
161 + ggplot() +
162 +   geom_polygon(data = us_states,
163 +               aes(x = long, y = lat, group = group),
164 +               fill = "gray95", color = "black", linewidth = 0.3) +
165 +   geom_point(data = tornado_paths |>
166 +             filter(elat == 0 | elon == 0),
167 +             aes(x = slon, y = slat),
168 +             color = "red",
169 +             alpha = 0.5,
170 +             size = 0.4) +
171 +   facet_wrap(~ mag, nrow = 3, ncol = 2) +
172 +
173 +   labs(
174 +     title = "Starting Location for Tornadoes with Missing End Coordinates Grouped by
175 +           Magnitude",
176 +     x = "Longitude",
177 +     y = "Latitude"
178 +   )
179 + ```
180 +
181 + :::
182 +
183 + ## How Will a Probability Be Computed? {.smaller}
184 +
8 185
9   - - Dividing sharknado probability into a sequence of necessary probabilities
10  - -
11 186   $$
12   - P(\text{sharknado | tornado is coastal to inland}) =
13 187   P(\text{shark is present}) \times
14 188   P(\text{tornado is strong enough to lift a shark | tornado is coastal to inland})
15   - $$
16   - }
189 + $$
190 +
191 + - We assume that  $P(\text{shark is present}) = 1$ 
192 +
193 + - If probability is still extremely low then sharknado risk will be essentially non existent
194 +
195 + - Classify each tornado as coastal to inland, inland to coastal, or neither
196 +
197 + | **Tornado Type** | **Starting Distance from Coast** | **Ending Distance from Coast** |
198 + | **Criteria** |

```

```

199 + | Coastal → Inland | ≤ 10,000 m | \> Starting distance | Moves away from coast |
200 + | Inland → Coastal | \> 10,000 m | \< Starting distance | Moves toward coast |
201 + | Other | Could be either | Could be either | Doesn't meet the 2 criteria for either group |
202 +
203 + ## <span style="font-size: 0.89em;">How Will a Probability Be Computed? cont.</span>
204 +
205 + ::: {style="text-align:center;"}
206 +
207 + 
208 +
209 + <br>
210 +
211 + <span style="color:red;">Coastal to Inland</span> |
212 + <span style="color:blue;">Inland to Coastal</span>
213 +
214 + :::
215 +
216 +
217 + ## Probabilities of Each Type of Tornado {.smaller}
218 +
219 + - simply divide the number of tornadoes in the respective category by the number of total
    tornadoes
220 +
221 + ```{r, echo = FALSE}
222 + library(rnaturalearth)
223 + library(sf)
224 +
225 + tornado_paths <- tornado_paths |>
226 +   filter(end_missing == FALSE)
227 +
228 + tornado_paths$start_geom_degrees <- st_as_sf(
229 +   tornado_paths,
230 +   coords = c("slon", "slat"),
231 +   crs = 4269
232 + )$geometry
233 +
234 + tornado_paths$end_geom_degrees <- st_as_sf(
235 +   tornado_paths,
236 +   coords = c("elon", "elat"),
237 +   crs = 4269
238 + )$geometry
239 +
240 + tornado_paths$start_geom_meters <- st_transform(
241 +   tornado_paths$start_geom_degrees, 5070
242 + )
243 +

```

```
244 + tornado_paths$end_geom_meters <- st_transform(
245 +   tornado_paths$end_geom_degrees, 5070
246 + )
247 +
248 + coast_m <- ne_coastline(returnclass = "sf") |>
249 +   st_transform(5070)
250 + # Find distances to coast and classify each tornado as coastal_to_inland , inland_to_coastal
    , or other
251 + tornado_paths <- tornado_paths |>
252 +   mutate(
253 +     dist_start_to_coast = st_distance(start_geom_meters, coast_m) |> apply(1, min),
254 +     dist_end_to_coast = st_distance(end_geom_meters, coast_m) |> apply(1, min)
255 +   )
256 +
257 + tornado_paths <- tornado_paths |>
258 +   mutate(
259 +     movement = case_when(
260 +       dist_start_to_coast <= 10000 & dist_end_to_coast > dist_start_to_coast ~
        "Coastal_to_Inland",
261 +       dist_start_to_coast > 10000 & dist_end_to_coast < dist_start_to_coast ~
        "Inland_to_Coastal",
262 +       TRUE ~ "Other"
263 +     )
264 +   )
265 + library(knitr)
266 + # Find probabilities of each type of tornado
267 + number_coastal_to_inland <- tornado_paths |>
268 +   filter(movement == "Coastal_to_Inland") |>
269 +   nrow()
270 +
271 + number_inland_to_coastal <- tornado_paths |>
272 +   filter(movement == "Inland_to_Coastal") |>
273 +   nrow()
274 +
275 + number_other <- tornado_paths |>
276 +   filter(movement == "Other") |>
277 +   nrow()
278 +
279 + total_tornadoes <- nrow(tornado_paths)
280 +
281 + probability_coastal_to_inland <- 100*(number_coastal_to_inland/total_tornadoes)
282 + probability_inland_to_coastal <- 100*(number_inland_to_coastal/total_tornadoes)
283 + probability_other <- 100*(number_other/total_tornadoes)
284 +
285 + tornado_df_for_table <- data.frame(
286 +   category = c("Coastal to Inland", "Inland to Coastal", "Other"),
```

```

287 +   number_of_tornadoes = c(number_coastal_to_inland, number_inland_to_coastal, number_other),
288 +   probability = c(probability_coastal_to_inland, probability_inland_to_coastal,
289 +   probability_other)
290 + )
291 + kable(
292 +   transform(
293 +     tornado_df_for_table,
294 +     probability = sprintf("%.2f%%", probability)
295 +   ),
296 +   col.names = c("Tornado Type", "Number of Tornadoes", "Probability")
297 + )
298 +
299 +
300 + ```
301 +
302 + - relatively small amount of coastal to inland tornadoes when compared to inland to coastal
303 +   and other
304 +
305 + - less amount of land close to the coast when compared to the interior of the country
306 +
307 + ## Assumptions for EF magnitude Needed For Sharknado {.smaller}
308 +
309 + - Great White Sharks can weigh up to 4,500 lbs
310 +
311 + - The average car weighs approximately 4,419 lbs
312 +
313 + - EF3 and below tornadoes have not been seen lifting cars significant distances
314 +
315 + - Only EF4 and EF5 tornadoes have had the ability to throw cars many feet
316 +
317 + {width=70%}
318 +
319 + ## Probability of a Sharknado {.smaller}
320 +
321 + - This restriction allows for only one possible sharknado out of the tornadoes we have left
322 +
323 + - A sharknado is only possible in  $\frac{1}{417} \approx 0.24\%$  of coastal to inland
324 +   tornadoes.
325 +
326 +

```

```

...   @@ -0,0 +1,234 @@
1     + ---
2     + title: "Results"
3     + format: html
4     + ---
5     +
6     + ```{r, echo = FALSE, message = FALSE, output = FALSE}
7     + library(gt)
8     +
9     + tornado_vars <- data.frame(
10    +   Variable = c("yr", "date", "st", "mag", "slat", "slon", "elat", "elon", "len", "wid"),
11    +   Description = c(
12    +     "Year Tornado Occurred",
13    +     "Exact Date Tornado Occurred",
14    +     "State Abbreviation Where Tornado First Touched Down",
15    +     "EF Scale Magnitude of the Tornado",
16    +     "Latitude Where the Tornado First Touched Down",
17    +     "Longitude Where the Tornado First Touched Down",
18    +     "Latitude Where the Tornado Ended",
19    +     "Longitude Where the Tornado Ended",
20    +     "Tornado Path Length (Miles)",
21    +     "Tornado Path Width (Yards)"
22    +   ),
23    +   Type = c("Integer", "Date", "Categorical", "Integer", "Numeric", "Numeric", "Numeric",
24    +     "Numeric", "Numeric", "Numeric"),
25    +   Possible_Values = c("1950-2024", "01-01-1950 to 12/31/2024", "Any US State", "0-5 or -9
26    +     Meaning Unknown", "US Latitudes ~ 29 to 50", "US Longitudes ~ -125 to -70", "US Latitudes ~
27    +     29 to 50", "US Longitudes ~ -125 to -70", "0-250 Miles", "0-3500 Yards")
28    + )
29    +
30    + tornado_vars %>%
31    +   gt() %>%
32    +     tab_header(
33    +       title = "Tornado Data Variable Descriptions"
34    +     ) %>%
35    +     cols_label(
36    +       Variable = "Variable Name",
37    +       Description = "Description",
38    +       Type = "Type",
39    +       Possible_Values = "Possible Values"
40    +     ) %>%
41    +     tab_style(
42    +       style = cell_text(weight = "bold"),
43    +       locations = cells_column_labels(everything())
44    +     ) %>%
45    +     tab_style(

```

```
43 +   style = cell_borders(
44 +     sides = "all",
45 +     color = "black",
46 +     weight = px(1)
47 +   ),
48 +   locations = cells_body(everything())
49 + ) %>%
50 +   tab_style(
51 +     style = cell_borders(
52 +       sides = "all",
53 +       color = "black",
54 +       weight = px(1)
55 +     ),
56 +     locations = cells_column_labels(everything())
57 +   )
58 + # Data Cleaning (1 row per tornado, only important columns, dealing with 0's in elat and
    elon)
59 + library(tidyverse)
60 +
61 + tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv") |>
62 +   filter(!st %in% c("AK", "HI", "PR", "VI"))
63 +
64 + tornado_paths <- tornado_full_data |>
65 +   distinct(slat, slon, elat, elon, .keep_all = TRUE)
66 +
67 + tornado_paths <- tornado_paths |>
68 +   select(slat, slon, elat, elon, mag)
69 +
70 + tornado_paths <- tornado_paths |>
71 +   mutate(end_missing = (elat == 0 | elon == 0))
72 +
73 + ggplot(tornado_paths, aes(x = slon, y = slat)) +
74 +   geom_point(color = "red", alpha = 0.5) +
75 +   labs(title = "Starting Location For All Tornadoes",
76 +        x = "Starting Longitude", y = "Starting Latitude") +
77 +   theme_minimal()
78 + ggplot(tornado_paths |> filter(elat == 0 | elon == 0),
79 +        aes(x = slon, y = slat)) +
80 +   geom_point(color = "red", alpha = 0.5) +
81 +   labs(title = "Starting Location For Tornadoes With Missing End Coordinates",
82 +        x = "Starting Longitude", y = "Starting Latitude") +
83 +   theme_minimal()
84 + library(ggtext)
85 + ggplot(
86 +   tornado_paths |>
87 +     filter(elat == 0 | elon == 0),
```

```
88 +   aes(x = slon, y = slat)
89 + ) +
90 +   geom_point(color = "red", alpha = 0.5) +
91 +   facet_wrap(~ mag) +
92 +   labs(
93 +     title = "Starting Location For Tornadoes With Missing End Coordinates (by Magnitude)",
94 +     x = "Starting Longitude",
95 +     y = "Starting Latitude"
96 +   ) +
97 +   theme_minimal() +
98 +   theme(
99 +     plot.title = element_markdown()
100 +   )
101 + # Transforming latitude and longitude into math ready quantities
102 + library(rnaturalearth)
103 + library(sf)
104 +
105 + tornado_paths <- tornado_paths |>
106 +   filter(end_missing == FALSE)
107 +
108 + tornado_paths$start_geom_degrees <- st_as_sf(
109 +   tornado_paths,
110 +   coords = c("slon", "slat"),
111 +   crs = 4269
112 + )$geometry
113 +
114 + tornado_paths$end_geom_degrees <- st_as_sf(
115 +   tornado_paths,
116 +   coords = c("elon", "elat"),
117 +   crs = 4269
118 + )$geometry
119 +
120 + tornado_paths$start_geom_meters <- st_transform(
121 +   tornado_paths$start_geom_degrees, 5070
122 + )
123 +
124 + tornado_paths$end_geom_meters <- st_transform(
125 +   tornado_paths$end_geom_degrees, 5070
126 + )
127 +
128 + coast_m <- ne_coastline(returnclass = "sf") |>
129 +   st_transform(5070)
130 + # Find distances to coast and classify each tornado as coastal_to_inland , inland_to_coastal
    , or other
131 + tornado_paths <- tornado_paths |>
132 +   mutate(
```



```
133 +   dist_start_to_coast = st_distance(start_geom_meters, coast_m) |> apply(1, min),
134 +   dist_end_to_coast = st_distance(end_geom_meters, coast_m) |> apply(1, min)
135 + )
136 +
137 + tornado_paths <- tornado_paths |>
138 +   mutate(
139 +     movement = case_when(
140 +       dist_start_to_coast <= 10000 & dist_end_to_coast > dist_start_to_coast ~
141 +         "Coastal_to_Inland",
142 +       dist_start_to_coast > 10000 & dist_end_to_coast < dist_start_to_coast ~
143 +         "Inland_to_Coastal",
144 +       TRUE ~ "Other"
145 +     )
146 +   )
147 + ::: callout
148 + introduce the section
149 + :::
150 +
151 + Now we need to compute the probabilities for each tornado type that we have classified. This
152 +   can be done quite simply by dividing the number of tornadoes that we have classified to each
153 +   tornado type by the total number of tornadoes. The results are summarized in the following
154 +   table:
155 +
156 + ```{r, echo = FALSE}
157 + library(knitr)
158 + # Find probabilities of each type of tornado
159 + number_coastal_to_inland <- tornado_paths |>
160 +   filter(movement == "Coastal_to_Inland") |>
161 +   nrow()
162 +
163 + number_inland_to_coastal <- tornado_paths |>
164 +   filter(movement == "Inland_to_Coastal") |>
165 +   nrow()
166 +
167 + number_other <- tornado_paths |>
168 +   filter(movement == "Other") |>
169 +   nrow()
170 +
171 + total_tornadoes <- nrow(tornado_paths)
172 +
173 + probability_coastal_to_inland <- 100*(number_coastal_to_inland/total_tornadoes)
174 + probability_inland_to_coastal <- 100*(number_inland_to_coastal/total_tornadoes)
175 + probability_other <- 100*(number_other/total_tornadoes)
```

```

174 + tornado_df_for_table <- data.frame(
175 +   category = c("Coastal to Inland", "Inland to Coastal", "Other"),
176 +   number_of_tornadoes = c(number_coastal_to_inland, number_inland_to_coastal, number_other),
177 +   probability = c(probability_coastal_to_inland, probability_inland_to_coastal,
178 +     probability_other)
179 + )
180 + kable(
181 +   transform(
182 +     tornado_df_for_table,
183 +     probability = sprintf("%.2f%%", probability)
184 +   ),
185 +   col.names = c("Tornado Type", "Number of Tornadoes", "Probability")
186 + )
187 +
188 +
189 + ```
190 +
191 + As shown, there are far fewer coastal-to-inland tornadoes compared with inland-to-coastal and
192 +   other tornadoes. This relatively small amount reflects the strict requirement that the
193 +   tornado must start within a short distance of the coast to be classified in this category.
194 +
195 + ```{r, echo = FALSE}
196 + #Find probabilities of each magnitude of each type of tornado
197 + options(scipen = 999)
198 +
199 + probabilities <- tornado_paths |>
200 +   group_by(mag, movement) |>
201 +   summarise(count = n(), .groups = "drop") |>
202 +   group_by(movement) |>
203 +   mutate(magnitude_proportion_given_movement_type = count / sum(count)) |>
204 +   ungroup() |>
205 +   mutate(proportion_of_total_tornadoes = count / total_tornadoes)
206 +
207 + probabilities_for_coastal_to_inland <- probabilities |>
208 +   filter(movement == "Coastal_to_Inland") |>
209 +   mutate(magnitude_probability_given_movement_type =
210 +     100*magnitude_proportion_given_movement_type)
211 + ```
212 +
213 + Lastly, we estimate the probability that a tornado is strong enough to lift a shark,
214 +   conditional on it being coastal-to-inland. According to @fisheriesWhiteSharkNOAA2025, great
215 +   white sharks can weigh up to 4,500 lbs. Similarly, @HowMuchDoes reports that the average car
216 +   weighs approximately 4,419 lbs, suggesting that great white sharks are of comparable mass to
217 +   cars. From @CanCarBe2025, EF3 tornadoes are “likely to toss cars and trucks,” whereas EF4
218 +   tornadoes are “very likely to toss cars many feet.” This indicates that at least an EF4

```

tornado would be required, since a sharknado would require sharks not only to be lifted, but also transported a substantial distance inland rather than merely displaced. To compute this probability, we first have to define the number of tornadoes for each combination of magnitude m and movement type t , call this $N_{m,t}$. The probability of a tornado having magnitude m given movement type t is the number of tornadoes with magnitude m and movement type t divided by the total number of tornadoes with movement type t .

211 +

212 + ::: callout

213 + also mention that they have to be pulled out of water – is that possible to calculate with water being an added effect?

214 + :::

215 +

216 + \$\$

217 + $P(\text{Magnitude} = m | \text{Movement} = t) = \frac{N_{m,t}}{\sum_m N_{m,t}}$

218 + \$\$ For our specific probability we are interested in this probability where $m = 4$ or 5 , and $t = \text{coastal to inland}$.

219 +

220 + Computing this probability we get:

221 +

222 + \$\$

223 + $\frac{\text{probabilities_for_coastal_to_inland}[6, \text{"count"}] + 0}{\sum(\text{probabilities_for_coastal_to_inland}[\text{count}])}$

224 + $\approx \text{round}(\text{probabilities_for_coastal_to_inland}[6, \text{"magnitude_probability_given_movement_type"}], 2) \%$

225 + $\quad$

226 + $(1 \text{ from magnitude } 4, 0 \text{ from magnitude } 5)$

227 + \$\$

228 +

229 + ::: callout

230 + double check how the math notation is interpreting your text visually since underscores turn the following letters or symbols into a subscript. Also what is the ending probability numerically other than converging to zero?

231 + :::

232 +

233 + It is important to note that even this estimate represents a conservative upper bound. We assume that a shark is present along the coast whenever a coastal-to-inland tornado occurs, and that an EF4 tornado is sufficient to lift a shark from the ocean onto land. These assumptions are deliberately favorable to the occurrence of a sharknado. Relaxing these large assumptions would cause this probability to converge even closer to 0.

234 +

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